Bulletin of the
Seismological Society of America

Vol. 70 February 1980 No. 1

A SPECTRAL THEORY FOR CIRCULAR SEISMIC SOURCES; SIMPLE ESTIMATES OF SOURCE DIMENSION, DYNAMIC STRESS DROP, AND RADIATED SEISMIC ENERGY

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ABSTRACT

The far-field body wave radiation from a class of circular rupture models is investigated as a function of takeoff angle, rupture velocity, and stopping behavior. In particular, the variation of spectral shape, pulse shape, and energy flux over the focal sphere is quantified. These results provide two new methods for estimating the source dimension, the first through the inversion of a characteristic frequency, and the second using the rise time of the displacement pulse shape. The class of kinematic rupture models which has been constructed also allows a direct estimate of the dynamic stress drop using the velocity pulse shape: as an example, this technique is applied to an SMA1 recording of a high stress drop Aleutian earthquake. Finally, the introduction of a new spectral parameter, the integral of the square of the ground velocity, is shown to considerably simplify calculation of the total radiated energy. Together with a re-evaluation of corner frequency and spectral falloff, these new results constitute a spectral theory for circular seismic sources which includes directivity and is valid for a range of subsonic rupture velocities.

Introduction

In his work on the spectral theory of seismic sources, Randall (1973) derived the scaling of seismic spectra with source size for a set of proposed spectral shapes by comparing the radiated energy (computed from the spectra) to the theoretical energy available for seismic radiation. This comparison determined a relation between corner frequency and source dimension without specifying a particular source model, vindicating Brune's (1970) results. There are, however, two simplifying assumptions in Randall's derivation which restrict the applicability of his results.

The first of these assumptions is that all of the available energy is radiated seismically. (The available energy is the strain energy released less the energy lost to frictional heating on the fault surface.) As Madariaga (1976) and Husseini (1977) have shown for general frictional models, the available energy is partitioned between the radiated energy and the fracture energy (i.e., the energy lost to the heating and plastic deformation accompanying the rupture growth.) The slower the rupture velocity, the less available energy is radiated. The limiting case is a quasi-static rupture which radiates no energy. Thus the scaling of seismic spectra with source size, as derived from the radiated energy, necessarily depends on the rupture velocity. This dependence has also been directly established for specified source models by a number of authors, notably Savage (1974) and Anderson and Richards (1975). Essentially, their results show that measurements of corner frequency or pulse width determine estimates of the rupture duration, i.e., the source dimension

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divided by the rupture velocity, rather than the source dimension. Only by estimating or assuming a value for the rupture velocity can the source dimension be obtained from the rupture duration.

The second assumption in Randall's derivation is that the spectral shape and the corner frequency are constant over the focal sphere. As we will show for rupture velocities greater than half the shear-wave velocity the pulse shapes and spectra vary strongly over the focal sphere even for simple circular ruptures, so that their interpretation will also depend on the takeoff direction of the ray. Since our sources are circular, this variation depends only on the scalar slowness of the body wave along the fault plane and describes only one aspect of the general directivity present in the body waves. The other aspect of body-wave directivity, i.e., the effects of a strongly asymmetrical rupture propagation, is beyond the scope of this study.

In order to quantify this directivity in the time domain, we introduce an analysis of the energy flux pulse shapes, as obtained from the time histories of the square of the ground velocity. Such v^2-plots were first used by Hanks (1974) in an analysis of the Pacoima Dam recording, and more recently by Boatwright (in preparation) in an analysis of a multiply recorded aftershock of the 1975 Oroville, California earthquake. For traces where the body-wave arrival contains the dominant energy, v^2-plots provide generally coherent pulse shapes which both clearly detail the body-wave directivity and also allow a simple estimate of the rupture duration.

The integral of these v^2 pulse shapes provides a useful parameter for further quantifying the seismic spectrum. By Parseval's theorem, the integral of the square of the ground velocity is equal to the second moment of the displacement power spectrum. Complementing the usual spectral parameters (namely, the low-frequency asymptote to the displacement spectrum and the corner frequency), this parameter simplifies the estimation of the radiated energy and also provides an estimate of the rupture duration through the definition of a characteristic frequency.

The purpose of this paper is to provide simple techniques for estimating source parameters from body-wave data for small and moderate sized earthquakes. Begin by deriving formulas for the seismic moment and the total radiated energy using a ray-theoretical representation theorem for a heterogeneous elastic medium. Since the relations between source radius, radiated energy, and the seismic spectrum are model dependent (specifically varying with rupture velocity and stopping character), we have analysed a class of simple kinematic rupture models which incorporate the results of Kostrov (1964) and Madariaga (1976), from which we obtain synthetic pulse shapes and spectra. The energy analysis for circular sources parallels Haskell's (1964) analysis for propagating line sources. The computation of synthetic pulse shapes is considerably simplified by the exploitation of the Fraunhofer approximation for finite sources, which is also used to quantify the relation between the P- and S-wave pulse shapes and spectra.

Representations for M_0 and E_s

The specific pulse shape and spectral analyses discussed in this study are based on the simple representation for the far-field ground motion due to body waves. Body-wave arrivals provide useful information about the seismic source owing to their relative lack of distortion by frequency-dependent wave propagation effects. The necessary incorporation of attenuation is discussed qualitatively rather than formally included in the representation.

The undistorted body-wave pulse shapes may be obtained by integrating the

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relative slip velocity $\Delta\dot{u}(\xi, t)$ (i.e., the velocity of one side of the fault relative to the other side),

$$\Omega_c(\theta, \phi, t) = \iint_\Sigma \Delta\dot{u}(\xi, t - T^c(\boldsymbol{x}, \xi))\, d\Sigma \tag{1}$$

where Σ is the rupture area and $T^c(\boldsymbol{x}, \xi)$ is the body-wave travel time from ξ (a specific point on Σ) to the receiver at $\boldsymbol{x}$. The source-receiver geometry is shown in Figure 1. We will refer to $\Omega_c(\theta, \phi, t)$ as the *model pulse shape.* Define the *model pulse area* as

$$\Omega_0 = \int_0^\infty \Omega_c(\theta, \phi, t)\, dt = \iint_\Sigma \Delta u(\xi, \infty)\, d\Sigma, \tag{2}$$

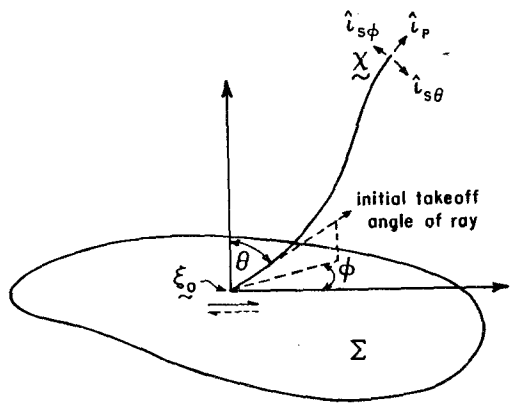

FIG. 1. Source-receiver geometry. Σ is the fault plane, θ is the takeoff angle, measured relative to the fault normal, and ϕ is the azimuth, measured relative to the direction of slip on the fault; $\hat{\imath}_p$, $\hat{\imath}_{s\theta}$ and $\hat{\imath}_{s\phi}$ define the ray coordinates for the p, $s\theta$, and $s\phi$ particle motion.

and note that the seismic moment is simply

$$M_0 = \mu(\xi_0)\Omega_0 \tag{3}$$

where $\mu(\xi_0)$ is an average rigidity for the hypocentral region.

To obtain the far-field body wave motions radiated by our sources, we use a ray-theoretical representation for the displacement of a heterogeneous elastic medium (Aki and Richards, 1979, Chapter 4),

$$\boldsymbol{u}_c(\boldsymbol{x}, t) = \frac{\mu(\xi_0)\hat{\imath}_c}{4\pi[\rho(\xi_0)\rho(\boldsymbol{x})c(\boldsymbol{x})]^{1/2}c(\xi_0)^{5/2}} \frac{F^c(\theta, \phi)}{R^c(\boldsymbol{x}, \xi_0)} \Omega_c(\theta, \phi, t). \tag{4}$$

Here $\rho(\xi_0)$ and $c(\xi_0)$ are the average density and wave velocity for the hypocentral region; $\rho(\boldsymbol{x})$ and $c(\boldsymbol{x})$, the density and wave velocity at the receiver; F^c (θ, ϕ), the radiation pattern coefficient of the body wave with takeoff angles θ and ϕ; $\hat{\imath}_c$, the direction at the receiver of the particle motion of the body wave; and $R^c(\boldsymbol{x}, \xi_0)$, the geometrical spreading factor. The notation c is used both for the vector of the

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particle motion whose basis is fixed to the ray ($c = p, s\theta, s\phi$), and for the wave type ($c = \alpha, \beta$). The coordinates θ and ϕ are fixed by the direction of slip on the fault. For simplicity, we assume the geometrical spreading factors for P and S waves are identical, i.e., $R^c(\boldsymbol{x}, \xi_0) \equiv R(\boldsymbol{x}, \xi_0)$.

These theoretical displacements may be compared to seismic body-wave data which has been corrected for the effects of attenuation and the free surface. For direct arrivals, the geometrical spreading factor fully describes the amplification of the propagation path. Relevant formulas for calculating $R(\boldsymbol{x}, \xi_0)$ may be found in Newman (1973) and Aki and Richards (1979, Chapter 4). For reflected arrivals, the geometrical spreading factor must be multiplied by the appropriate reflection coefficient. Throughout the rest of this section, we will use $u_c(\boldsymbol{x}, t) = \boldsymbol{\varepsilon}_c \cdot \boldsymbol{u}_c(\boldsymbol{x}, t)$, and $\bar{u}_c(\boldsymbol{x}, \omega)$ for the Fourier transform of $u_c(\boldsymbol{x}, t)$.

To estimate the moment of an event from body-wave data, re-write equation (4) and integrate over time to obtain

$$M_0 = 4\pi[\rho(\xi_0)\rho(\boldsymbol{x})c(\boldsymbol{x})]^{1/2}c(\xi_0)^{5/2}\frac{R(\boldsymbol{x}, \xi_0)}{F^c(\theta, \phi)}\bar{u}_c(\boldsymbol{x}, 0). \tag{5}$$

Here $\bar{u}_c(x, 0)$ is the displacement pulse area, or equivalently, the low-frequency asymptote of the displacement spectrum. Thus, the ray-theoretical estimate of the moment depends on the receiver properties of density and wave velocity as well as the usual source properties, radiation pattern, and geometrical spreading.

To obtain a similar result for E_s, the radiated seismic energy, we need to extend the representation theorem to consider the energy flux in body waves. From Bullen (1965) the energy flux rate (or power flux) through a surface at $\boldsymbol{x}$ due to a normally incident plane wave of velocity $c(\boldsymbol{x})$ is

$$\dot{\epsilon}_c(\boldsymbol{x}, t) = \rho(\boldsymbol{x})c(\boldsymbol{x})\dot{u}_c^2(\boldsymbol{x}, t). \tag{6}$$

This shows that the power flux varies exactly as the square of the ground velocity, for seismograms with separate P and S arrivals. The energy flux may be obtained by integrating this power flux over time:

$$\epsilon_c(\boldsymbol{x}) = \int_0^\infty \dot{\epsilon}_c(\boldsymbol{x}, t)\, dt = \rho(\boldsymbol{x})c(\boldsymbol{x})I_c(\boldsymbol{x}). \tag{7}$$

Here $I_c(x)$ is the integral of the square of the ground velocity. By Parseval's relation,

$$I_c(\boldsymbol{x}) = \int_0^\infty \dot{u}_c^2(\boldsymbol{x}, t)\, dt = \frac{1}{\pi}\int_0^\infty \omega^2 |\bar{u}_c(\boldsymbol{x}, \omega)|^2\, d\omega. \tag{8}$$

In calculating this parameter from severely band-limited data, corrections for the finite bandwidth (resulting either from the instrument or from noise contamination) should be included.

Defining the *model pulse energy* as

$$\Psi_c(\theta, \phi) = \int_0^\infty \dot{\Omega}_c^2(\theta, \phi, t)\, dt, \tag{9}$$

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substitution of equation (4) into equations (8) and (7) determines the radiated energy flux at $\boldsymbol{x}$;

$$\epsilon_c(\boldsymbol{x}) = \left(\frac{\mu(\xi_0)F^c(\theta, \phi)}{4\pi R(\boldsymbol{x}, \xi_0)}\right)^2 \frac{\Psi_c(\theta, \phi)}{\rho(\xi_0)c^5(\xi_0)}. \tag{10}$$

To evaluate the total radiated seismic energy, E_s, integrate this energy flux for all three components of motion over a sphere containing the source. Taking this integral over the focal sphere for simplicity gives

$$E_s = \int_0^{2\pi} \int_0^{\pi} R^2(\boldsymbol{x}, \xi_0)[\epsilon_p(\boldsymbol{x}) + \epsilon_{s\theta}(x) + \epsilon_{s\phi}(\boldsymbol{x})] \sin\theta \, d\theta \, d\phi. \tag{11}$$

Substitution of equation (10) into equation (11) gives

$$E_s = \left(\frac{\mu(\xi_0)}{4\pi}\right)^2 \frac{\Psi_0}{\rho(\xi_0)} \tag{12}$$

where we have defined the *model integrated energy*, ψ_0, as

$$\Psi_0 = \int_0^{2\pi} \int_0^{\pi} \left[F^{P^2}(\theta, \phi) \frac{\psi_\alpha(\theta, \phi)}{\alpha^5(\xi_0)} + [F^{s\theta^2}(\theta, \phi) + F^{s\phi^2}(\theta, \phi)] \frac{\psi_\beta(\theta, \phi)}{\beta^5(\xi_0)}\right] \sin\theta \, d\theta \, d\phi. \tag{13}$$

Finally, in order to estimate the radiated energy from a measurement of $I_c(x)$, divide equation (12) by equation (10) and use equation (7) to obtain

$$E_s = \left(\frac{R(\boldsymbol{x}, \xi_0)}{F^c(\theta, \phi)}\right)^2 \frac{c^5(\xi_0)\Psi_0}{\Psi_c(\theta, \phi)} \rho(\boldsymbol{x})c(\boldsymbol{x})I_c(\boldsymbol{x}). \tag{14}$$

Defining a dimensionless *fractional energy* flux as

$$e_c(\theta, \phi) \equiv \frac{\Psi_c(\theta, \phi)}{c^5(\xi_0)\Psi_0} \tag{15}$$

simplifies the estimate of the radiated energy to

$$E_s = \left(\frac{R(\boldsymbol{x}, \xi_0)}{F^c(\theta, \phi)}\right)^2 \frac{\rho(\boldsymbol{x})c(\boldsymbol{x})}{e_c(\theta, \phi)} I_c(\boldsymbol{x}). \tag{16}$$

In previous investigations of the radiated energy (Hanks and Wyss, 1972; Randall, 1973), E_s was determined as a function of both the moment and the corner frequency. The parameter $I_c(x)$, however, gives a more direct formulation since it does not require estimating other parameters. Assuming the spectral shape to be constant over the focal sphere, Randall obtained $e_\beta(\theta, \phi) \approx 1/2\ \pi$ and $e_\alpha(\theta, \phi) = \beta^2(\xi_0)/3\pi\alpha^2(\xi_0)$. However, as will be shown in the section on Radiated Energy, the fractional

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energy flux varies strongly with the takeoff angle for the models analyzed, and the variation itself depends on the rupture velocity.

Circular Quasi-Dynamic Models

In order to investigate the model dependence involved in the inversion for source dimension and radiated energy, we have analyzed the body waves from a class of circular source models. In these models the region of stress relaxation grows with a uniform rupture velocity until the rupture front either decelerates continuously or stops abruptly. In the first model, the interior of the rupture area heals as the rupture front decelerates. In the second, the rupture front stops abruptly, and the interior heals in a causal fashion. The stopping and healing of these models is azimuthally symmetric and therefore lacking in generality. However, by considering both instantaneous and causal healing behaviors, it is hoped a range of results that are relevant to realistic healing behaviors may be established. The "self-similar" slip distribution analyzed by Kostrov (1964) and Burridge and Willis (1969) was used to describe the rupture growth of these models. Because this rupture growth is associated with a dynamically feasible behavior of the shear stress on the fault plane, these models are referred to as quasi-dynamic.

The first source model grows from a point with a uniform rupture velocity until a time t_s when the rupture front begins to decelerate, slowing continuously until time t_h when the entire rupture area heals. This model will be referred to as the Decelerating or D-model.

The relative slip displacement is specified in three distinct phases: the self-similar rupture growth,

$$\Delta u_d(r, t) = \Delta \dot{u}_0 \left(t^2 - \frac{r^2}{v^2}\right)^{1/2} \qquad \frac{r}{v} \leqq t \leqq t_s; \quad (17a)$$

the deceleration,

$$\Delta u_d(r, t) = \Delta \dot{u}_0 (t^2 - \upsilon\left(t - t_s)^2 - \frac{r^2}{v^2}\right)^{1/2} \qquad t_s < t < t_h \quad (17b)$$

and finally, the static offset,

$$\Delta u_d(r, t) = \Delta \dot{u}_0 \frac{a}{v} \left(1 - \frac{r^2}{a^2}\right)^{1/2} \qquad t \geqq t_h. \quad (17c)$$

In these equations $r = |\xi - \xi_0|$, v is the initial rupture velocity, a is the final fault radius, and $\Delta \dot{u}_0$ is a particle velocity which depends on the dynamic stress drop; υ is a parameter which determines the duration of the deceleration. The times t_s and t_h are given by

$$t_s = \frac{a}{v} \left(\frac{\upsilon - 1}{\upsilon}\right)^{1/2} \quad (18a)$$

$$t_h = \frac{a}{v} \left(\frac{\nu}{\upsilon - 1}\right)^{1/2}. \quad (18b)$$

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For finite v, the D-model satisfies the constraints enumerated by Dahlen (1974), so that the body wave spectra are expected to show falloffs of ω^{-3} in the high frequencies. In the limit as $v \rightarrow \infty$, this model becomes identical to that used by Sato and Hirasawa (1973), whose pulse shapes and spectra are easily derived analytically, providing us with a useful check of our numerical procedure. Note that this S&H model stops instantaneously at time $t_s = t_h = a/v$.

Figure 2 shows the relative slip velocity at five different radii for the D-model with $t_s = 0.82\ a/v$. The half-order singularity at the rupture front has been smoothed over $0.025\ a/v$. Note the deceleration of the rupture front (the delay before the rupture reaches $r = 0.95\ a$) and the acausal onset of healing over the entire rupture area at time t_h.

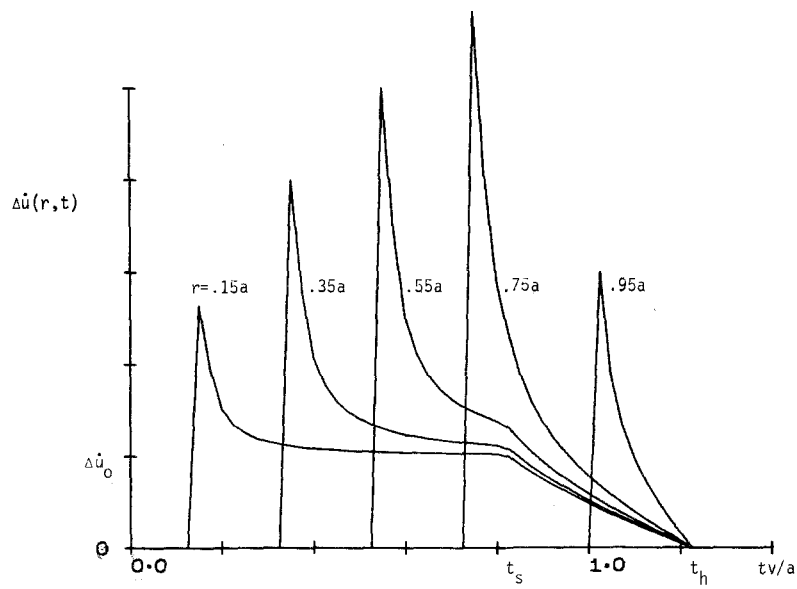

FIG. 2. Plots of the relative slip velocity, $\Delta\dot{u}(r, t)$, for the D-model, at five different radii. The slip is self-similar until $t_s = 0.82\ a/v$ when the fault begins to decelerate. The slip heals over the entire fault surface at $t_n = 1.22\ a/v$. Note the delay of the rupture front arrival for $r = 0.95\ a$.

The particle velocity, $\Delta\dot{u}_0$ which scales the relative slip displacement of these quasi-dynamic models is given by Kostrov (1964) and Dahlen (1974) as

$$\Delta\dot{u}_0 = C\left(\frac{v}{\beta}\right)\frac{\tau e}{\mu(\xi_0)}\beta(\xi_0) \approx \frac{v\tau_e}{\mu(\xi_0)} \tag{19}$$

where τ_e, the dynamic stress drop (or the effective stress, Brune, 1970) is the difference between the initial (tectonic loading) stress and the dynamic frictional stress, and $C(v/\beta)$ is the Kostrov function (Dahlen, 1974). For these models, $C(0.6) = 0.59$, $C(0.75) = 0.71$, $C(0.9) = 0.82$. Our notation is determined from the result that $\Delta\dot{u}(0, t) = \Delta\dot{u}_0$ during the rupture growth.

The formulation of these kinematic models has two important features: first, the body-wave pulse shapes scale directly with the dynamic stress drop, and second, there is no need for rise time as a source parameter. In this model, the final slip distribution is obtained by simply stopping the dynamic solution, which yields a uniform static stress drop over the rupture area (Eshelby, 1957).

To derive the moment in terms of the source parameters of the model, integrate

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equation (17c) over the fault area and substitute equation (19):

$$M_0 = \frac{2\pi}{3}\mu(\xi_0)\Delta\dot{u}_0\frac{a^3}{v} \approx \frac{2\pi}{3}\tau_e a^3. \tag{20}$$

The ratio of the static stress drop to the dynamic stress drop may be obtained by substituting the relation

$$M_0 = \frac{16}{7}\Delta\sigma a^3 \tag{21}$$

(Brune, 1970) for the left hand side of equation (20):

$$\frac{\Delta\sigma}{\tau_e} = \left(\frac{7\pi}{24}\right)\frac{C\left(\frac{v}{\beta}\right)}{\frac{v}{\beta}}. \tag{22}$$

Because the D-models heal as soon as the rupture front reaches the final perimeter, they have the smallest stress ratio possible for a source with the self-similar rupture growth and unreversed slip. Note that all the D-models have the same final distribution of slip.

The second model we have constructed has abrupt stopping, but, in contrast to the D-model, has causal healing. The rupture grows with a uniform rupture velocity until reaching a fixed rupture perimeter where it stops abruptly. The motion at an interior point continues in the rupture growth phase until the P-wave generated by the stopping of the rupture front arrives at the interior point at time $t_s(r)$. As the rupture heals, the relative slip velocity decreases continuously to zero. Since this model essentially reproduces Madariaga's (1976) results from his numerical model of dynamic stress relaxation, we refer to it as the Madariaga, or M-model.

The relative slip displacement of this model differs from the D-model only in the generalization of the times t_s and t_h to functions of radius; therefore the rupture growth is similar, i.e.,

$$\Delta u_m(r, t) = \Delta\dot{u}_0\left(t^2 - \frac{r^2}{v^2}\right)^{1/2} \qquad \frac{r}{v} < t < t_s(r), \tag{23a}$$

as is the form of the healing (or deceleration),

$$\Delta u_m(r, t) = \Delta\dot{u}_0\left(t^2 - v(r)[t - t_s(r)]^2 - \frac{r^2}{v^2}\right)^{1/2} \qquad t_s(r) < t < t_h(r) \tag{23b}$$

while the static offset differs by the factor ρ_0,

$$\Delta u_m(r, t) = \Delta\dot{u}_0\frac{a\rho_0}{v}\left(1 - \frac{r^2}{v^2}\right)^{1/2} \qquad t > t_h(r). \tag{23c}$$

The causal healing of this model begins at time $t_s(r)$, while the definitions of $t_n(r)$

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and $v(r)$ ensure a uniform static stress drop;

$$t_s(r) = \frac{a}{v} + \frac{(a - r)}{\alpha} \tag{24a}$$

$$t_h(r) = \frac{\rho_0^2 a^2 + (1 - \rho_0^2) r^2}{v^2 t_s(r)} \tag{24b}$$

$$v(r) = \frac{t_h(r)}{t_h(r) - t_s(r)}. \tag{24c}$$

The factor ρ_0 determines the ratio of the final slip of these models to the final slip

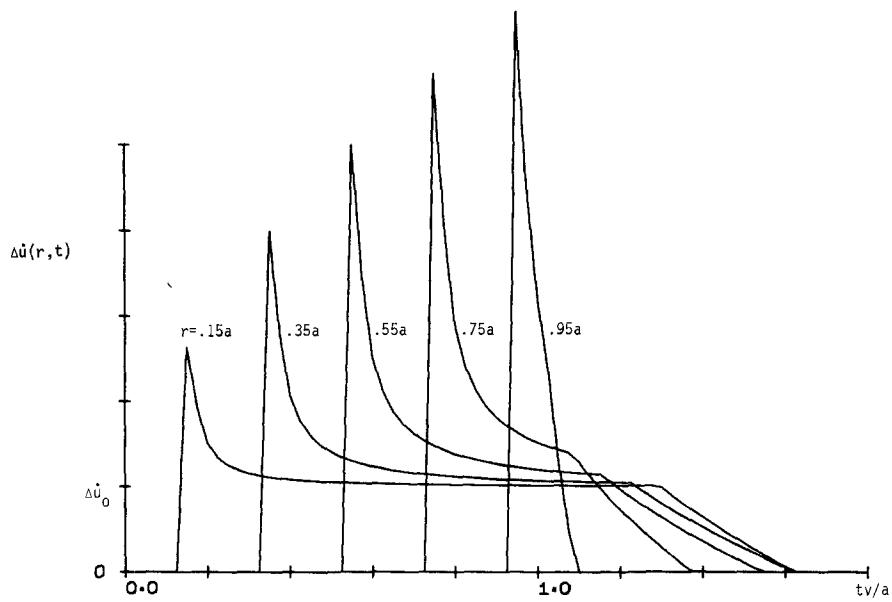

FIG. 3. Plots of the relative slip velocity, $\Delta\dot{u}(r, t)$, for the M-model with $v = 0.9\ \beta(\xi_0)$, at five different radii. The motion at each radius is self-similar until $t_s(r)$, the arrival time of the P-wave stopping phase from the perimeter of the fault. The corners in the slip velocity mark this arrival time.

of the D-model, i.e.,

$$\Delta u_m(r, \infty) = \rho_0 \Delta u_d(r, \infty). \tag{25}$$

Since the static offset of the M-model is simply ρ_0 times that of the D-model, the moment and the stress ratio of the M-model may be obtained by multiplying the right hand sides of equations (20) and (22) by ρ_0.

Figure 3 shows the relative slip velocity at five different radii for the M-model with $v = 0.9\ \beta(\xi_0)$. The arrival of the P-wave stopping phase determines the onset of the healing, producing a distinct corner in the relative slip velocity. Note the different healing behavior at the different radii.

In contrast to the D-model, the M-model has a substantial overshoot. Thus in the two models there is both an extreme range of stopping and healing behavior, and also a large range of the stress ratio, owing to the contrast of $\rho_0 = 1$ for the D-model and $\rho_0 \cong 1.5$ for the M-model.

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The Fraunhofer Approximation

The far-field pulse shapes radiated by these models is calculated using the Fraunhofer approximation for the travel time,

$$T^c(\boldsymbol{x}, \xi) = T^c(\boldsymbol{x}, \xi_0) - \frac{|\xi - \xi_0|}{c(\xi_0)} \sin\theta \cos\phi'. \tag{26}$$

(Aki and Richards, 1980, Chapter 14) in the integrand of equation (1). Here ϕ' is the azimuth of the ray relative to the velocity $\xi - \xi_0$, as measured on the fault plane. The resulting integral

$$\Omega_c[\theta, \phi, t + T^c(\boldsymbol{x}, \xi_0)] = \int\int_\Sigma \Delta\dot{u}(\xi, t + \frac{|\xi - \xi_0|}{c(\xi_0)} \sin\theta \cos\phi')\, d\Sigma \tag{27}$$

is numerically evaluated on a radial grid centered at ξ_0 using the time derivatives of the slip distribution defined in equations (17) and (23).

To ensure that this numerical integration gives a stable result, the half-order singularity in the relative slip velocity at the rupture front ($r_e = vt$) has been analytically integrated over the distance to the grid point at $r_e - \lambda$. The approximation,

$$\int_{r_e-\lambda}^{r_e} \Delta\dot{u}(r, \tau) r\, dr = \Delta\dot{u}_0 \int_{r_e-\lambda}^{r_e} \left(\tau^2 - \frac{r^2}{v^2}\right)^{-1/2} \tau r\, dr$$

$$\approx \Delta\dot{u}_0 r_e \left(\frac{\lambda}{1 - \zeta\cos\phi'}\right)^{1/2} [2r_e - \lambda(1 - \zeta\cos\phi')]^{1/2}, \tag{28}$$

determines the contribution to the integral of equation (27) from the strip of width λ at the rupture front. In these equations, the definitions $\theta \equiv v \sin\theta/c(\xi_0)$ and $\tau \equiv t + \zeta r \cos\phi'/v$ have been used.

The relative error of the Fraunhofer approximation increases with frequency. By allowing the error in the travel time to be at most a quarter of the wave period, a high-frequency limit for the approximation can be determined; in the notation of the first two sections, this limit is

$$\nu_{\max} \approx T^c(\boldsymbol{x}, \xi_0) \frac{c^2(\xi_0)}{2a^2} \tag{29}$$

(Aki and Richards, 1980, Chapter 14). Thus, the frequency range where the approximation is valid depends on both the travel time and the source size.

Despite this high-frequency limit, the Fraunhofer approximation provides valuable insight into the directivity in the far-field pulse shapes. In particular, note that for a specific source the integrand of equation (27) is a function of $\sin\theta \cos\phi'/c(\xi_0)$, showing that the directivity in the far-field pulse shapes depends only on the vector slowness of the ray on the fault plane. As Kostrov (1970) pointed out, the P-wave pulse shape at θ_p will be identical to the S-wave pulse shape at

$$\theta_s = \sin^{-1}\left(\frac{\beta(\xi_0)}{\alpha(\xi_0)} \sin\theta_p\right) \tag{30}$$

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if both rays take off at the same azimuth. This result implies that the P-wave pulse shapes are a subset of the S-wave pulse shapes.

For circular models whose time dependence may be written as a function of $t^* = vt/a$ (for example, the D-model), the pulse shapes may be exactly parameterized by the variable

$$\zeta = \frac{v}{c(\xi_0)} \sin \theta, \tag{31}$$

which will be called the *apparent Mach number*. To demonstrate this result, define the dimensionless distance, $r^* = r/a$; then the time plus the travel-time delay in the integrand of equation (27) may be written as

$$\tau = \frac{a}{v} (t^* + r^* \zeta \cos \phi'). \tag{32}$$

Because the time dependence of the rupture growth phase (equations 17a and 23a) is a function of t^*, equation (32) indicates that our pulse duration measurements should vary approximately as a/v, while our spectral measurements should vary as v/a. We refer to the quantity a/v as the *rupture duration*. The apparent Mach number is the natural independent variable. Note that a specific upper bound for ζ determines an upper bound for the takeoff angle θ and thus describes a cone of takeoff angles around the normal to the fault plane.

Spectral Falloff and Corner Frequency

The variation of spectral shape over the focal sphere has been theoretically investigated by a number of authors, notably Savage (1974), Dahlen (1974), and Madariaga (1976, 1977). The present analysis is concerned with both the identification of intermediate-frequency trends and the general variation of γ, the rate of spectral falloff, for spectra generated by models which exhibit a range of stopping behaviors.

Figure 4 shows displacement pulse shapes and velocity spectra from the far-field radiation of the D-model with $t_s = 0.82\ a/v$. The displacement pulse shapes are normalized as $\Omega_c'(\theta, \phi, t) = \Omega_c(\theta, \phi, t)/(\Delta \dot{u}_0 a^2)$ and plotted as functions of dimensionless time. As shown in Figure 5, where measurements of the rate of spectral falloff for different versions of the D-model have been compiled, any significant duration of deceleration produces ω^{-3} falloffs within the range $\zeta < 0.5$. In the figure annotation we use $r_s = vt_s$. The S&H model, in which $r_s = a$, provides the necessary control, as it has an equivalent stress ratio but stops abruptly and shows ω^{-2} falloffs.

If the log-log model spectra had more than one distinguishable linear trend, both falloff rates were plotted. For $\zeta < 0.6$, the D-model develops an intermediate slope region with $1.2 < \zeta < 1.7$, which becomes more pronounced for higher ζ, as may be seen in Figure 4. This behavior was first pointed out by Savage (1974). It occurs because the rise time of the displacement pulse shape decreases with increasing ζ. At $\zeta = 1$, the rise becomes discontinuous, and the spectral falloff becomes ω^{-1}.

Figure 6 shows the pulse shapes and spectra from the M-model, which have been analyzed in order to assess the effects of causal healing. This healing behavior has little effect other than to increase the overshoot, producing somewhat fatter pulse shapes. Figure 7 further demonstrates that an abrupt stopping behavior gives ω^{-2} falloffs in the range $\zeta < 0.5$. For large ζ, the M-model spectra have intermediate slope regions whose falloff rates vary as $1.4 < \gamma < 1.7$.

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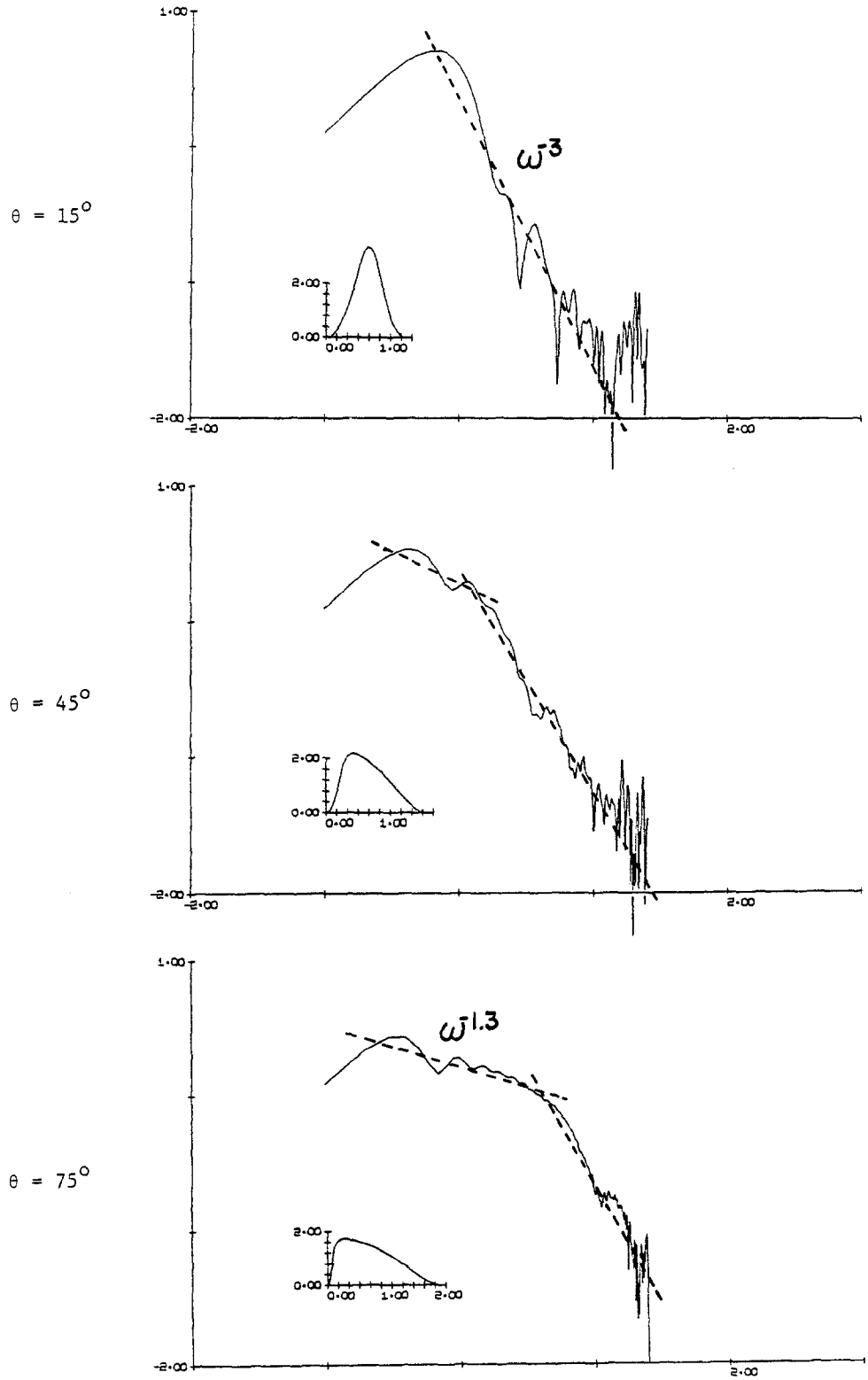

FIG. 4. Displacement pulse shapes and log-log velocity spectra from the D-model with $t_s = 0.82\ a/v$ at takeoff angles of $\theta = 15°$, 45°, and 75° for $v = 0.9\ \beta(\xi_0)$. The displacement pulse shapes are plotted as functions of normalized time, vt/a, while the spectra are functions of normalized frequency, $a\nu/v$. The velocity spectra was used to more clearly identify intermediate frequency trends. The spectral falloffs refer to the falloff of the *displacement* amplitude spectra. Note the variation of the falloff from ω^{-3} to $\omega^{-1.3}$.

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The corner frequency estimates were obtained from these spectra by fitting a straight line through the first distinguishable linear trend as a mean rather than as an envelope, since the spectral peaks (in actual data) are statistically less reliable than the means. Because of the generally band-limited nature of seismic data, the true asymptotic means are not always recoverable, so if an intermediate frequency trend was sufficiently well developed in the synthetic spectra, it was used to estimate the corner frequency using the intersection with the low-frequency asymptote.

In Brune's (1970) theory, the corner frequencies of shear waves vary as $\beta(\xi_0)/a$.

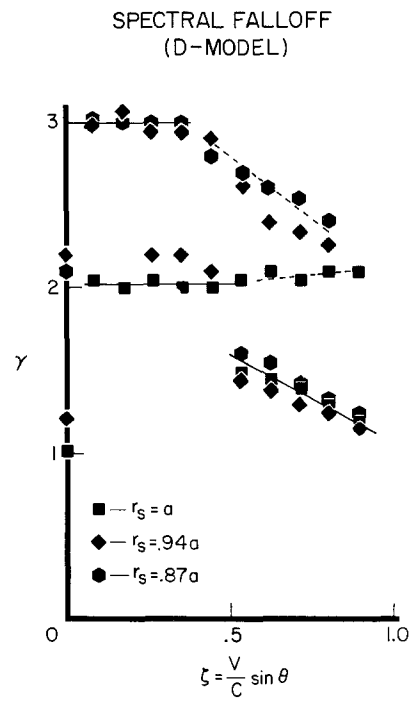

FIG. 5. Spectral falloff as a function of $\zeta = v \sin \theta/c$ for the D-model results. Where an intermediate and a high-frequency falloff are measured from the same spectrum, the same symbol is used. The dashed lines are drawn through the high-frequency falloffs from these spectra.

Generalizing this variation so that it also depends on the takeoff angle gives the relation

$$k_c(\theta) = \frac{a\nu_c(\boldsymbol{x})}{c(\xi_0)}, \tag{33}$$

where $\nu_c(\boldsymbol{x})$ is the corner frequency measured at $\boldsymbol{x}$. Because of the arguments presented at the end of the last section, the variation of $a\nu_c(x)/v$ over the focal sphere will be analyzed. It will be found, however, that equation (33) better describes the variation of the corner frequency for the range $\zeta < 0.6$.

The corner frequency measurements for all the models in Figure 8 have been compiled. As the M-models differ very slightly with rupture velocity in this scaling, the differences between their corner frequencies are small. The D-model corner frequencies show a great deal of scatter in the range $\zeta < 0.5$, and are substantially higher than the M-model corner frequencies. This difference results partly from the faster falloffs of the D-models, and partly from the difference in overshoot. The

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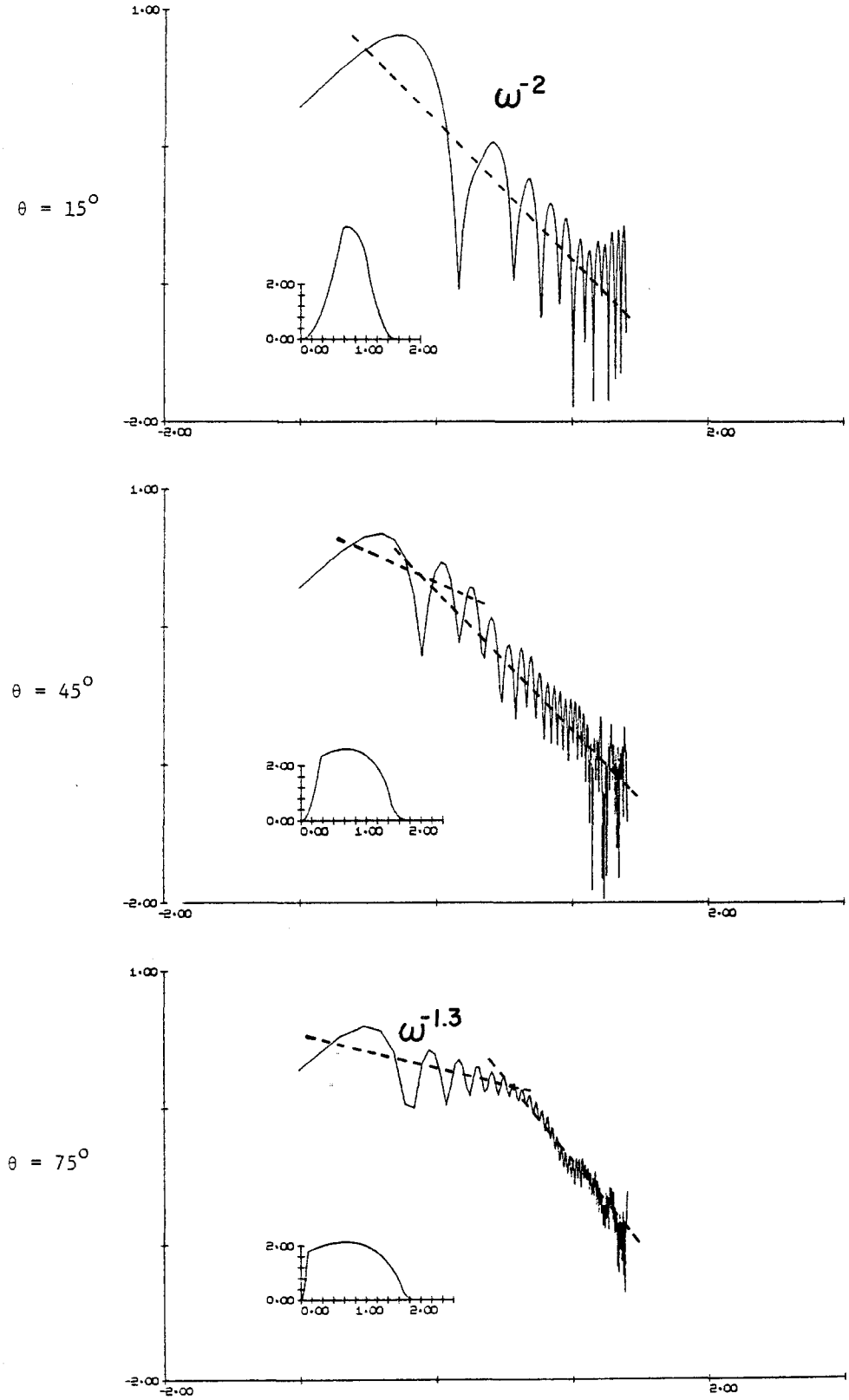

FIG. 6. Displacement pulse shapes and velocity spectra from the M-model with $v = 0.9\,\beta(\xi_0)$ at takeoff angles of 15°, 45°, and 75°. Note the smaller variation of the spectral falloff and the fatter pulse shapes than in the D-model results in Figure 4.

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effect of the stress ratio is demonstrated by the difference between the S&H model results and the M-model results.

Figure 8 also plots Dahlen's (1974) theoretical relation for corner frequencies from spectra which fall off as ω^{-3}

$$\frac{\nu_c a}{v} = \frac{\left(\frac{3}{4\rho_0}\right)^{1/3}}{\pi(1-\zeta^2)^{2/3}}. \tag{34}$$

This result was derived from his equations (39) and (40), using equations (19) and (20) from this paper. For the line in the figure, $\rho_0 = 1$. It appears that the high-frequency radiation of the D-model is not controlled by Dahlen's nucleation-dependent corner frequency.

Using the intermediate slope region to determine the corner frequency for $\zeta < 0.6$

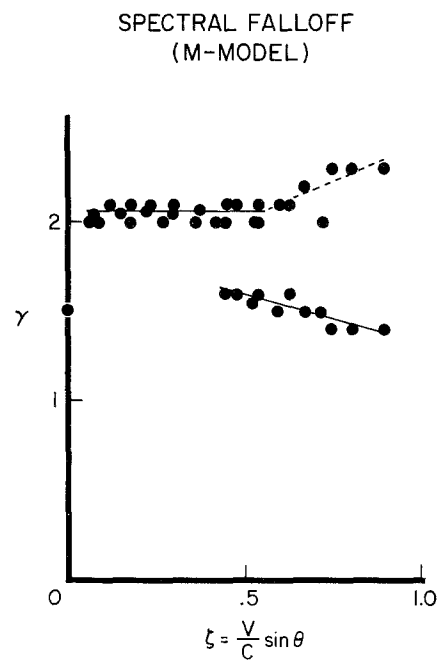

FIG. 7. Spectral falloff as a function of ζ for the M-model. Since the M-models with different rupture velocities differ only slightly (in their healing rate), the spectral falloffs are similar for similar ζ.

produces slight discontinuities in the plots of $a\nu_c(x)/v$ against ζ. The line drawn through the corner frequency for $\zeta > 0.6$ is the hyperbola

$$\frac{\nu_c(x)a}{v} = \frac{0.16}{\zeta}.$$

Writing this equation as a function of the takeoff angle gives the relation

$$\frac{\nu_c(x)a}{c(\xi_0)} = \frac{0.16}{\sin\theta} = k_c(\theta) \tag{35}$$

for the corner frequencies of shear waves radiated along the plane of the fault. Since

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this inversion is independent of the rupture velocity (see Madariaga, 1976, Figure 10), a direct estimate of the source dimension may be obtained. However, an intermediate slope region which falls off approximately as $\omega^{-1.5}$ must be used to determine the corner frequency from this relation (see Boatwright, 1978). Moreover, the event must be simple so that the intermediate slope resulting from a complex rupture (Boore and Joyner, 1978) is not mistaken for the intermediate slope in Figures 4 and 6.

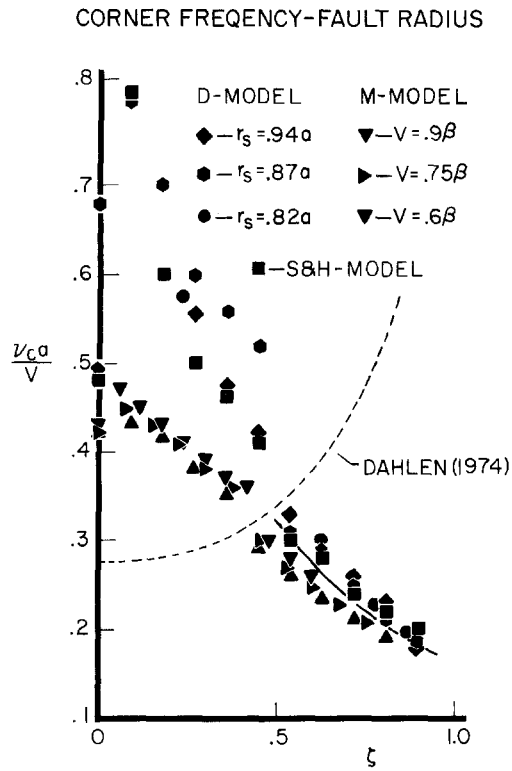

FIG. 8. Corner frequency as a function of ζ for all models. Since the M-models differ only slightly, the three M-models (▼, ▶, ▲) define one curve. Results for the D-models (◆, ⬢, ●) are extremely scattered in the range $\zeta < 0.5$. The line drawn through the corner frequency estimates for $\zeta > 0.6$ is the hyperbola $\nu_c(x)a/v = 0.16\ \zeta$. Dahlen's theoretical curve [see text, equation (34)] is also plotted, and is seen to be in complete disagreement.

CHARACTERISTIC FREQUENCY

For a specific spectral shape, the measurements of the low-frequency asymptote to the displacement spectrum, the corner frequency, and the integral of the square of the ground velocity make up an overdetermined system, such that any of these spectral parameters may be determined from the other two. Considering the variability of the spectral shape and the corner frequency for different takeoff angles, it is reasonable to try to combine the other two spectral measurements to obtain an estimate of the rupture duration which does not require measuring the corner frequency.

Dividing the integral of the square of the ground velocity by the square of the low-frequency asymptote, from equations (3), (5), (7), and (10), we have

$$\frac{I_c(x)}{\bar{u}_c^{\,2}(x, 0)} = \frac{\Psi_c(\theta, \phi)}{\Omega_0^{\,2}}. \tag{36}$$

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From equations (3), (20), and (42), this quantity varies as

$$\frac{\Psi_c(\theta, \phi)}{\Omega_0^2} \propto \left(\frac{v}{a}\right)^3.$$

From the cube root of this relation, we define

$$\kappa_c(\theta) = \frac{I_c(\boldsymbol{x})}{(\bar{u}_c^2(\boldsymbol{x}, 0))}^{1/3} \frac{a}{v} = \eta_c(\boldsymbol{x}) \frac{a}{v}. \tag{37}$$

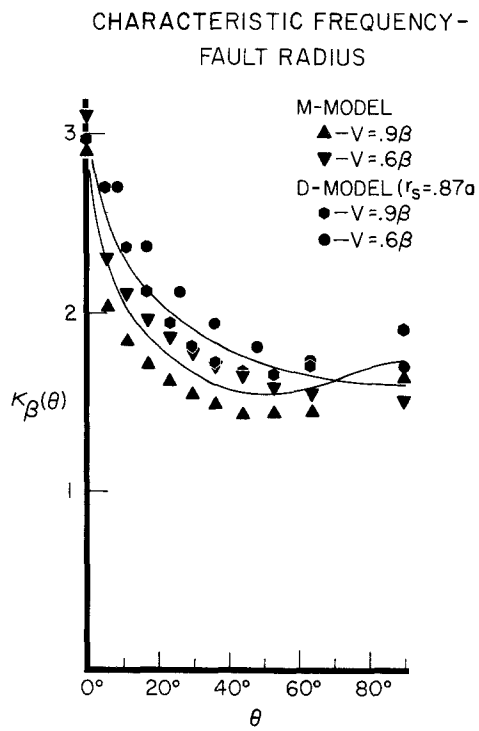

FIG. 9. Characteristic frequency as a function of takeoff angle for the M-model (▲, ▼) and the D-model (●, ●) with $r = 0.87\ a$. We have plotted measurements of $\kappa_\beta(\theta) = \eta_\beta(x)a/v$. The solid lines are estimates of $\kappa_\beta(\theta)$ for $v = 0.6$ and $0.9\ \beta(\xi_0)$, presuming a slip overshoot of $\rho_0 = 1.3$.

The quantity $\eta_c(\boldsymbol{x})$ has units of frequency and scales as expected for spectral parameters. Since this estimate requires an integration of the spectral shape function (Randall, 1973), we call $\kappa_c(\theta)$ the *shape parameter* and $\eta_c(\boldsymbol{x})$ the *characteristic frequency*.

The variation of the shape parameter over the focal sphere is shown in Figure 9, for both the M-model and the D-model with $r_s = 0.87\ a$, using two rupture velocities. It is interesting to note how the dependence on the overshoot and the rupture velocity is decoupled, so that $\kappa_c(\theta)$ varies with the slip overshoot approximately as $\rho_0^{-1/2}$ and varies with the rupture velocity as a function of the apparent Mach number. In this figure, curves have been drawn for $\kappa_c(\theta)$ for a slip overshoot of $\rho_0 = 1.3$, as a mean between the two models.

Outside the cone $\theta < 30°$, this method of estimating the rupture duration is nearly independent of the takeoff angle; over this range, $\langle \kappa_c(\theta) \rangle = 1.7$, where the bracket denotes the average for $30° < \theta < 90°$.

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Energy Flux Pulse Shapes

Energy flux pulse shapes, determined from the time history of the square of the ground velocity (v^2-plots) can provide useful modeling constraints as well as a simple estimate of the rupture duration. Figure 10 shows the S-wave v^2-plots radiated by

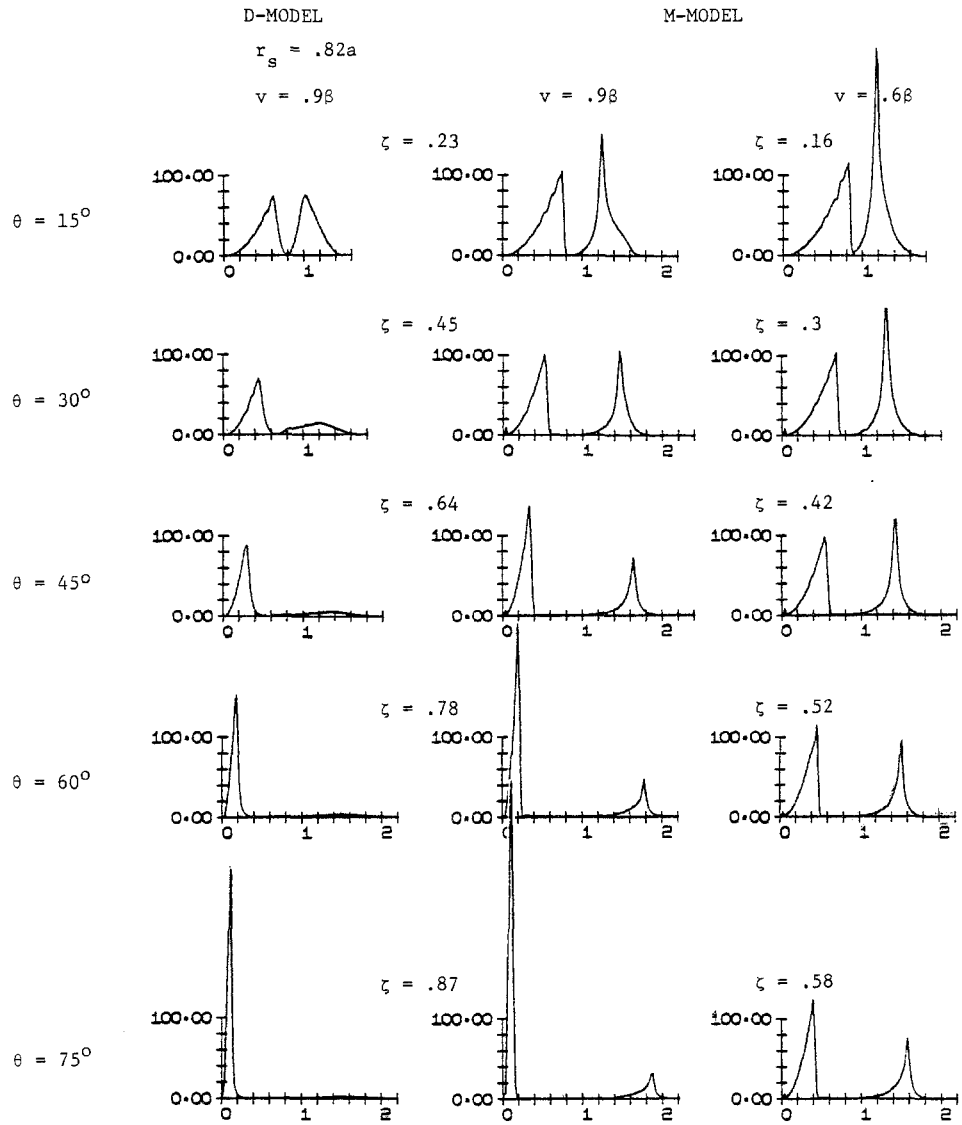

FIG. 10. S-wave v^2-plots from the source models. Note the overall behavior of these pulse shapes as natural functions of ζ. The sharp peaks of the M-model v^2-plots are an artifact of the healing specifications (see text).

our kinematic models at five different takeoff angles. Far-field velocity pulse shapes from simple sources are made up of one positive pulse and one negative pulse with equal areas. The resulting v^2-plots then have two separate pulses with generally unequal areas. The first pulse contains the energy flux from the growth of the rupture, while the second pulse contains the energy flux from the healing. In discussing velocity and v^2-plots, these two pulses are referred to as the rupture phase and the healing phase, respectively.

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The relative energy content of the rupture and healing phases varies substantially over the focal sphere. This is a result of the differences in the particle velocity behavior of the rupture front and during healing (see Figures 2 and 3). Within the cone $\theta < 30°$, where the healing of these models has the strongest constructive

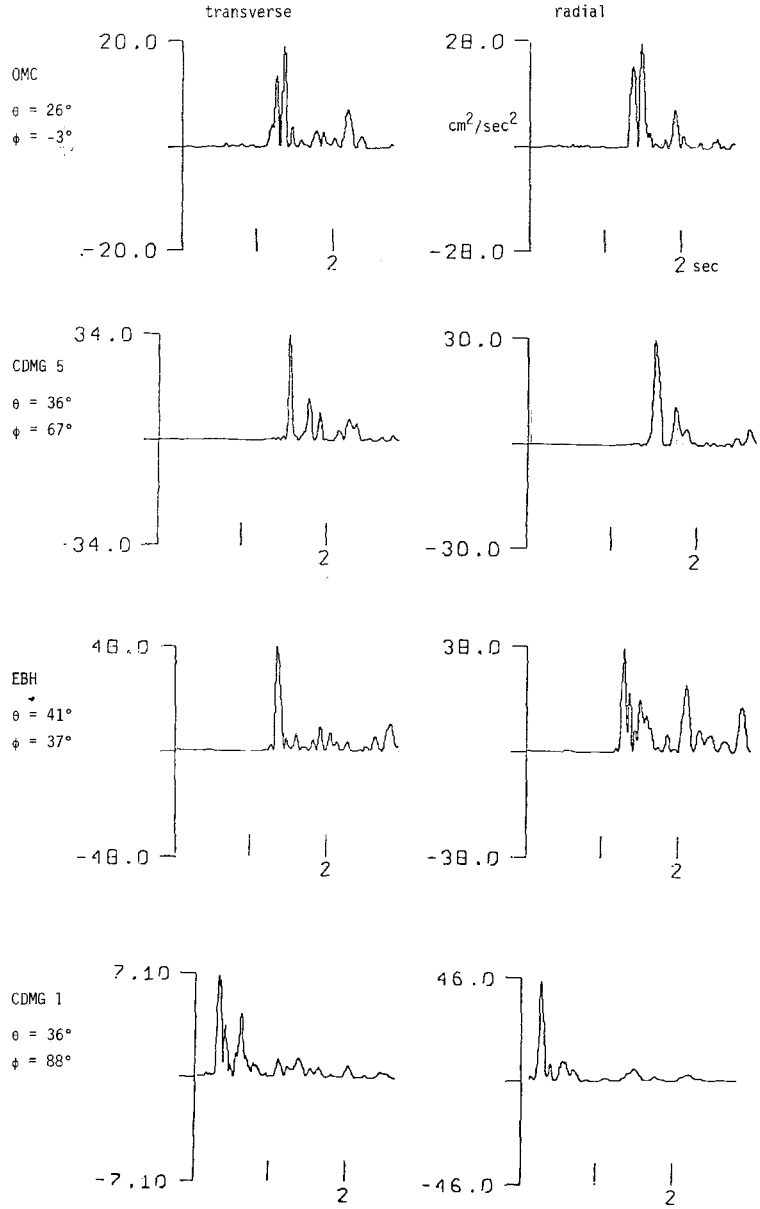

FIG. 11. S-wave v^2-plots from the Oroville aftershock data of Boatwright (1978). The transverse and radial horizontal components of motion are plotted for each station, along with the takeoff angle and azimuth of the ray relative to the slip direction.

interference, the two phases have approximately the same energy and pulse width. As ζ increases, the rupture phase becomes narrower and therefore more energetic. For circular sources, the relative amplitude of the phases varies monotonically with ζ; an asymmetrical rupture growth will strongly distort this behavior, however.

Figure 11 shows some of the Oroville aftershock v^2-plots from Boatwright (in

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preparation). Both the radial and transverse horizontal components of the S-wave arrivals have been plotted at four of the stations. As the rupture had a substantial asymmetry, these v^2-plots cannot be compared directly to the synthetic plots of Figure 10. The v^2-plots of Figure 11 show, however, an equivalent directivity which is clearly detailed by the v^2-plots.

The peaked behavior of the M-model healing phases is an artificial result of the large overshoot and the perfectly symmetric healing. For simple ruptures with less coherent stopping and healing, individual stopping events will dominate the high-frequency content of the healing phase, as in the Oroville v^2-plots.

The width of the rupture phase can be used to measure the displacement *pulse rise time*, $\tau_{1/2}$ (Bakun *et al.*, 1978). This pulse rise time, which should not be confused with the rise time characterizing the slip function (Haskell, 1964), permits a simple estimate of the source duration, as it gives the time delay between the first arrival and the arrival of the first stopping phase. From equation (26) this time delay may be calculated as $(1 - \zeta)a/v$. In Figure 12, $v\tau_{1/2}$ has been plotted against the apparent

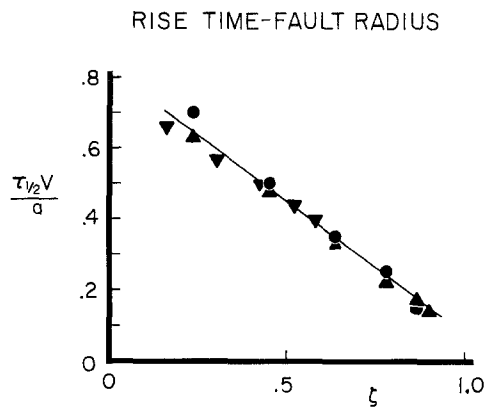

FIG. 12. Pulse rise time as a function of ζ, as measured directly from Figure 10 for the M-model (▲, ▼) and the D-model (●). The line drawn through the measurements is given by equation (38).

Mach number for the v^2-plots of Figure 10. Despite the substantial differences in stopping behavior between the two models, the pulse rise times show a consistent behavior which may be parameterized as

$$\tau_{1/2} \approx \frac{(13 - 12\zeta)}{16}\left(\frac{a}{v}\right). \tag{38}$$

Note that these rise times are obtained from the measurable width of the v^2 rupture phase, so that they are slightly smaller than the theoretical rise times.

Using measurements of the pulse rise times from the Oroville data in Figure 11 and assuming the rupture velocity is $0.85\beta(\zeta_0)$, an estimate of $a = 2.4 \pm 0.5$ km is obtained, which is somewhat large for this event. Assuming a unilateral rupture updip in the slip direction, and using $\tau_{1/2} \approx (1 - \zeta \cos \phi)a/v$ to invert the pulse rise time, where ϕ is the azimuth of the ray relative to the slip direction, we obtain $a = 1.4 \pm 0.1$ km.

DYNAMIC STRESS DROP

For the source models analyzed in this paper, the natural scaling of the slip functions in equations (17) and (23) by the factor $\Delta\dot{u}_0$ and the relation between $\Delta\dot{u}_0$

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and τ_e in equation (19) permits a simple estimate for the dynamic stress drop. We will derive two formulas for this estimate, the first applicable to any rupture having the self-similar rupture growth, and the second specific to model fits using the quasi-dynamic models.

The analytic form for the rupture phase of the velocity pulse shape radiated by the S&H model is

$$\dot{\Omega}_c[\theta, \phi, t + T^c(\boldsymbol{x}, \xi_0)] = \frac{4\pi\Delta\dot{u}_0 v^2 t}{(1 - \zeta^2)^2} \tag{39}$$

(Sato and Hirasawa, 1973). Since this result is derived only from the rupture growth and is unaffected by the stopping behavior, it represents a general form. Combining this with equations (4) and (19) gives a relation between the dynamic stress drop and the initial slope of the rupture phase of the velocity pulse shape

$$\tau_e = \frac{[\rho(\xi_0)\rho(\boldsymbol{x})c(\boldsymbol{x})]^{1/2}c(\xi_0)^{5/2}}{C\left(\frac{v}{\beta}\right)v^2\beta(\xi_0)}\left(\frac{R(\xi_0, \boldsymbol{x})}{F^c(\theta, \phi)}\right)(1 - \zeta^2)^2\left[\frac{\dot{u}_c(\boldsymbol{x}, t)}{t}\right]. \tag{40}$$

Here the term in brackets is the average initial slope. In Figure 13, where the application of these techniques to some SMA1 data from an Aleutian earthquake is shown, the dashed line on the velocity trace was used to estimate this average slope. Note that in equations (39) and (40) the factor $(1 - \zeta^2)^2$ contains the directivity affecting this calculation.

A similar estimate may be obtained through modeling velocity or v^2 pulse shapes using synthetics generated by the quasi-dynamic models. The amplitude of the data relative to the synthetics provides the estimate of the dynamic stress drop. The v^2-plots of Figure 10 are scaled as

$$\dot{\Omega}_c'^2(\theta, \phi, t) = \frac{\dot{\Omega}_c^2(\theta, \phi, t)}{(\Delta\dot{u}_0 va)^2}.$$

Fitting the synthetics to the data determines the factor

$$\left[\frac{\dot{u}_c(\boldsymbol{x}, t)}{\dot{\Omega}_c'(\theta, \phi, t)}\right]$$

which represents the average amplitude of the ground velocity relative to the dimensionless synthetic pulse shapes. The bracket indicates the average is taken over the whole pulse shape. Using equations (4) and (19), the estimate of the dynamic stress drop becomes

$$\tau_e = \frac{[\rho(\xi_0)\rho(\boldsymbol{x})c(\boldsymbol{x})]^{1/2}c(\xi_0)^{5/2}}{C\left(\frac{v}{\beta}\right)\beta(\xi_0)va}\left(\frac{4\pi R(\xi_0, \boldsymbol{x})}{F^c(\theta, \phi)}\right)\left[\frac{\dot{u}_c(\boldsymbol{x}, t)}{\dot{\Omega}_c'(\theta, \phi, t)}\right]. \tag{41}$$

Both of these formulas may be simplified by the approximation $C(v/\beta) \cong v/\beta(\xi_0)$, where $C(v/\beta)$ is the Kostrov function. Figure 13 also shows the synthetic fit to the v^2-plot for the Aleutian data. The synthetic pulse shape is the dashed line. For this

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event, the dynamic stress drop was 780 ± 250 bars, while the static stress drop, estimated using equation (21) and the source dimension techniques discussed in the last three sections, was 680 ± 200 bars (House and Boatwright, 1980).

These inversions provide direct estimates of the dynamic stress drop. Because of their strong dependence on the rupture velocity, these estimates have a relative standard error similar to estimates of the static stress drop. Note, however, that in equation (40) there is no dependence on estimates of source dimension, rise time or average dislocation, as in Kanamori (1972) and Boore and Stierman (1976).

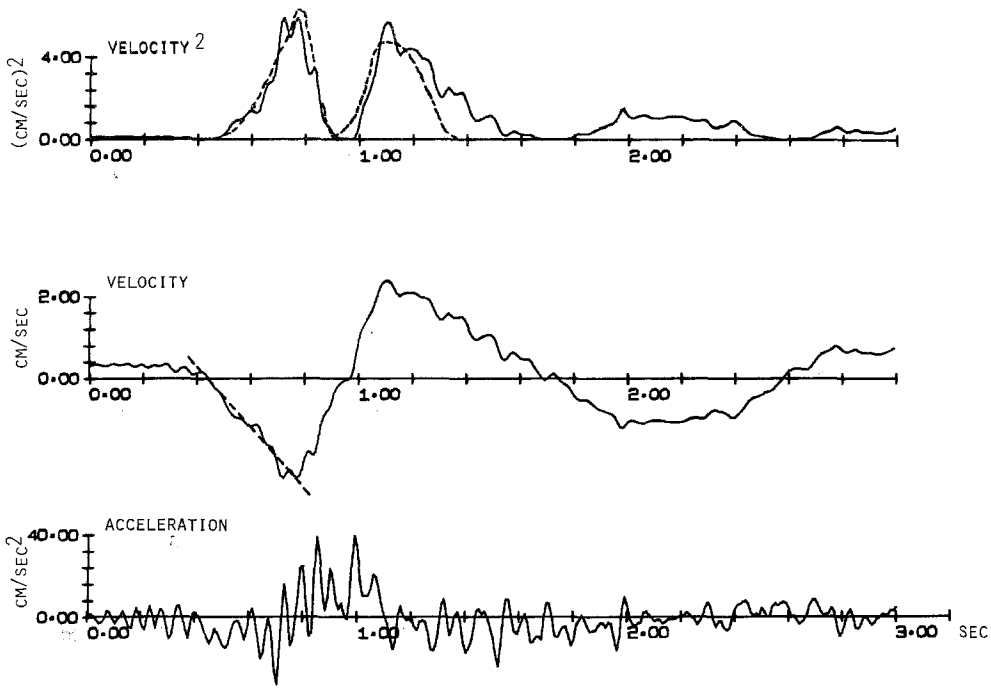

FIG. 13. Techniques for estimating the dynamic stress drop, as applied to the SV pulse shape from a 1974 Aleutian event. The velocity trace shows the estimation of the average slope used in equation (40). The uppermost trace is the v^2-plot along with the dashed synthetic used in equation (41). (Reprinted from House and Boatwright, 1980.)

RADIATED ENERGY

In order to calculate the seismic energy radiated by our synthetic sources, equation (13) must be integrated numerically, i.e., the total radiated energy must be obtained from the integral over the focal sphere of the energy flux contained in all three directions of particle motion. To evaluate this integral, the reciprocal of the model pulse energy, $\Psi_\beta(\theta)$, was fit with a fourth order polynomial in sin θ. Motivation for this interpretation comes from the analytic expression for $\Psi_c(\theta)$ from the Sato and Hirasawa model, where

$$\Psi_c(\theta) = \frac{2}{3}\pi^2 \Delta \dot{u}_0^2 v a^3 \left\{ \frac{3-\zeta}{\zeta(1-\zeta)(1+\zeta)^2} \right\}. \tag{42}$$

$\Psi_\alpha(\theta)$ was then obtained using equation (30), and these functions were integrated numerically over the focal sphere weighted by the wave velocities and the radiation patterns.

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This calculation for the radiated energy was checked against the theoretical relation,

$$E_s \approx \frac{\pi}{3} \frac{\tau_e^2}{\mu(\xi_0)} a^3 \left\{ \rho_0 \left(\frac{2 - \Delta\sigma}{\tau_e} \right) - g\left(\frac{v}{\beta} \right) \right\}, \tag{43}$$

[Madariaga, 1976, equation (32)], which was derived for circular ruptures which have uniform rupture velocities, abrupt stress drops, and unreversed slip. The relation was determined from the energy balance for frictional rupture models, where the radiated energy is simply the available energy less the fracture energy. The function $g(v/\beta)$ represents the relative amount of available energy given up as fracture energy. From Madariaga (1976), $g(v/\beta)$ varies monotonically from $g(0.9) = 0.21$ to $g(0.6) = 0.72$. The energy radiated by the kinematic sources agrees with this theoretical estimate to within 10 per cent. This agreement proves that the

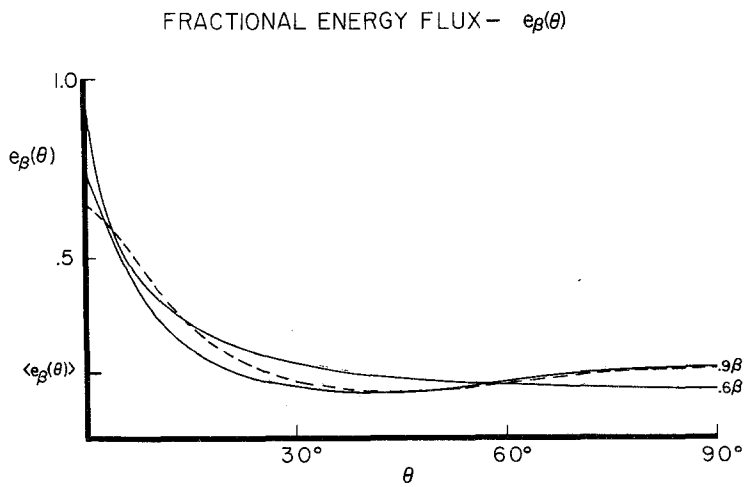

FIG. 14. The fractional energy flux for S-waves calculated for our models. The dashed line is for the D-model with $r_s = 0.87\ a$ and $v = 0.9\ \beta(\xi_0)$. The solid lines are for the M-model using $v = 0.6$ and 0.9 $\beta(\xi_0)$. $\langle e_\beta(\theta) \rangle$ was obtained by averaging $e_\beta(\theta)$ over the focal sphere, weighted by the radiation patterns.

quasi-dynamic models are energetically consistent with general theoretical models of dynamic rupture.

How the energy flux varies over the focal sphere was investigated by calculating the fractional energy flux for both P and S waves from equation (15). Figure 14 shows the results for $e_{\beta(\theta)}$ for three of the models. The most striking feature of this compilation is that the directions near the normal to the fault plane carry a relatively large amount of energy flux. This anomalous energy flux results from the constructive interference of the healing of the rupture, as seen in these directions, and the contribution of the healing phase to the total energy flux. While this is in part an artificial result of the symmetry of these models, since it is essentially a focusing effect it is reasonable to expect a maximum near $\theta = 0$ for any simple rupture which has an approximately equidimensional shape. The OMC v^2-plots in Figure 11 show an equivalent focusing which has been distorted by the asymmetrical rupture propagation to $\approx 30°$ from the fault normal in the direction of the rupture propagation.

For fast rupture velocities $[v > 0.85\beta(\xi_0)]$ the S-wave fractional energy flux has a

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local maximum for takeoff angles along the plane of the fault. As the rupture velocity approaches the shear-wave velocity this maximum becomes more pronounced. The average fractional energy flux for S waves, obtained by averaging the model pulse energy over the focal sphere weighted by the radiation patterns, is $\langle e_\beta(\theta)\rangle \approx 0.18$.

Figure 15 shows the P-wave fractional energy flux for the same models; $e_\alpha(\theta)$ decreases monotonically with increasing ζ for all models and rupture velocities. Following the calculation described above, we obtain $\langle e_\alpha(\theta)\rangle \approx 0.014$. For this class of models then, the ratio of the P-wave energy to the S-wave energy is 0.078; approximately $\frac{1}{13}$th of the seismic energy is radiated in P waves.

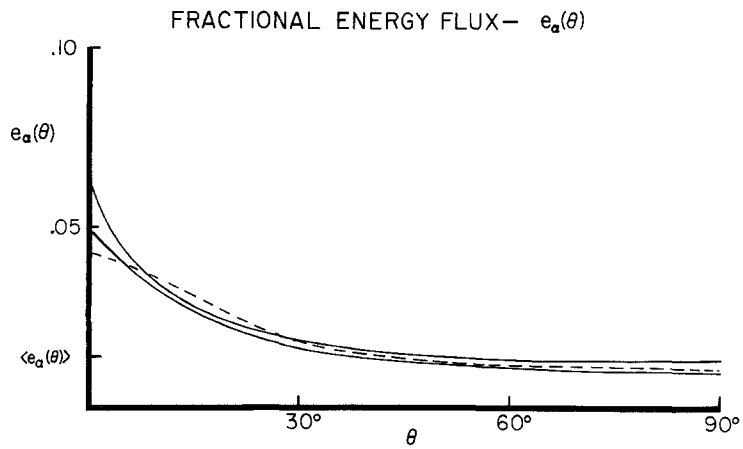

FIG. 15. The fractional energy flux for P waves for the same models used for Figure 14.

DISCUSSION AND CONCLUSIONS

In cataloging the far-field body-wave radiation in terms of an extended set of observational parameters, we have demonstrated the implicit dependence of source-dimension estimates on the rupture velocity. This dependence results both from the natural scaling of these parameters by the rupture duration (a/v) and from the approximate parameterization of the pulse shapes by the variable $\zeta = v \sin \theta/c$, which we have called the apparent Mach number. We have also demonstrated, however, that the corner frequencies of shear waves radiated near the plane of the fault may be used to give direct estimates of the source dimension if an intermediate slope is present in the spectrum.

The dependence on rupture velocity appears to introduce a new variable into the inversion for source parameters which would make these techniques less useful to the observational seismologist. However, since the range of observed rupture velocities is relatively small (generally considered to be between 0.6 and 0.9 times the shear-wave velocity), assuming a rupture velocity of $v = 0.75\ \beta(\xi_0)$ would reduce the uncertainty in this parameter to within 20 per cent.

The three techniques for estimating the source dimension (or rupture duration) in this paper, i.e., the corner frequency, the characteristic frequency, and the pulse rise time, should be applied together to reduce the inversion error. While the corner frequency is generally the weakest technique, it may provide the only direct estimate of the source dimension. The characteristic frequency can be used to estimate the rupture duration if the spectral shape cannot be simply interpreted or if the rupture is complex, in which case equation (37) gives an upper bound for the average

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subsegment rupture duration. The pulse rise time may be used to interpret velocity and v^2 pulse shapes directly and to check the spectral techniques. The simultaneous estimation of source size and rupture velocity, using a juxtaposition of these estimates, is not possible for data that does not sufficiently span the focal sphere. Because of this result, the rupture velocity acts as a hidden variable in the analysis of inadequate data.

The estimates of static and dynamic stress drop both have a strong (third-order) dependence on the rupture velocity. In estimating the dynamic stress drop, this dependence is explicit, as equations (40) and (41) show, while for the static stress drop, the dependence is implicit in the determination of the source dimension, as described above. The inversion for the dynamic stress drop formulated in equation (40) represents the most direct stress estimate possible as the only model uncertainty is the rupture velocity. This estimate strongly complements the usual estimate of the static stress drop; it is suggested that a combination of the two stress estimates should help reduce the general scatter of stress release estimates.

The radiation from actual three-dimensional ruptures will be naturally more complex than the synthetic pulse shapes from these simple source models. In particular, an asymmetrical rupture will generally distort the directivity described for these circular sources. While we have not modeled realistically incoherent stopping and healing, the healing behaviors used are presumed to describe adequate bounds for the low-frequency character of the healing phase from equidimensional ruptures, as the D-model heals very rapidly and the M-model heals slowly. As in the Oroville aftershock data, specific stopping events will generally determine the high-frequency character of the healing phase, unless the rupture stops in a very gradual fashion.

Acknowledgments

Over the course of this research I have had valuable discussions with Joe Andrews, Ralph Archulata, Dave Boore, Joe Fletcher, Tom Hanks, and Paul Richards. I am especially indebted to Tom Hanks for introducing me to the use of the v^2-plots and to Linda Lee Murphy who has now herded this manuscript through six drafts. The paper was critically reviewed by Paul Richards, Klaus Jacob, Dave Boore, and an unknown reviewer for the *Bulletin*, all of whose comments helped me enormously. This research was supported by the Advanced Research Projects Agency of the Department of Defense and was monitored by the Air Force Office of Scientific Research under Contract F49620-77-C-0098.

Glossary of Symbols

$\Omega_c(\theta, \phi, t)$ or $\Omega_c(\theta, t)$	— Model pulse shape
Ω_0	— Model pulse area
$F^c(\theta, \phi)$	— Radiation pattern coefficient
$R(\boldsymbol{x}, \xi_0)$	— Geometrical spreading factor
$T^c(\boldsymbol{x}, \xi_0)$	— Travel time
$\dot{\epsilon}_c(\mathbf{x}, t)$	— Energy flux
$\epsilon_c(\boldsymbol{x})$	— Radiated energy per unit area
$I_c(\boldsymbol{x})$	— Integral of squared ground velocity
$\Psi_c(\theta, \phi)$ or $\Psi_c(\theta)$	— Model pulse energy
Ψ_0	— Model integrated energy
$e_c(\theta, \phi)$ or $e_c(\theta)$	— Fractional energy flux
ρ_0	— Slip ratio
ζ	— Apparent Mach number
$k_c(\theta)$	— Corner frequency parameter
$\nu_c(\boldsymbol{x})$	— Corner frequency

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$\xi_f(\phi)$	— Fault perimeter
$\kappa_c(\theta)$	— Shape parameter
$\eta_c(c)$	— Characteristic frequency
$\tau_{1/2}$	— Displacement pulse rise time
$C\left(\frac{v}{\beta}\right)$	— Kostrov function
$g\left(\frac{v}{\beta}\right)$	— Fracture energy function

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Manuscript received May 22, 1978

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